

# Autoclaving

Autoclaving uses saturated steam under pressure (typically 121 °C, 15 psi) to sterilize media, buffers, glassware, and biohazardous waste. This protocol covers routine sterilization of laboratory items in a gravity-displacement steam autoclave, including pre-cycle preparation, cycle selection, and safe unloading.

## Reagents

| ✓                        | Name                                                            | Quantity         | Location/Notes                                      |
|--------------------------|-----------------------------------------------------------------|------------------|-----------------------------------------------------|
| <input type="checkbox"/> | Items to be sterilized (media, buffers, water, glassware, tips) | as needed        | Solids and liquids should be run on separate cycles |
| <input type="checkbox"/> | Distilled or deionized water (for water-bath fill if required)  | fill line        | Check chamber water level before each run           |
| <input type="checkbox"/> | Autoclave indicator tape                                        | 1 strip per item | Black diagonal stripes appear when sterile          |

## Equipment

| ✓                        | Name                                         | Quantity   | Location/Notes                                          |
|--------------------------|----------------------------------------------|------------|---------------------------------------------------------|
| <input type="checkbox"/> | Gravity-displacement or pre-vacuum autoclave | 1          |                                                         |
| <input type="checkbox"/> | Autoclave bags (polypropylene, biohazard)    | as needed  | Do NOT seal tightly — leave open for steam penetration  |
| <input type="checkbox"/> | Autoclave-safe secondary containment tray    | 1 per load | Catches spills from liquids; required for liquid cycles |
| <input type="checkbox"/> | Heat-resistant gloves and lab coat           | 1 set      | Required PPE - chamber surfaces remain hot after cycle  |

## Instructions

### Pre-cycle preparation

1. Inspect the autoclave: confirm the chamber is clean, the door gasket is intact, and the chamber water reservoir is filled to the indicator line.
2. Check the autoclave logbook for the most recent biological-indicator validation. If validation is more than one month old, run a spore-vial validation cycle before use.
3. Prepare items as listed in the Reagent Tracker below. Loosen all caps so steam can equilibrate inside containers, sealed containers may shatter or fail to sterilize.
4. Affix autoclave indicator tape across the opening of every bag, bottle, or wrapped item. Label each item with contents, date, and your initials using a permanent marker.
5. For liquids, transfer all containers into a secondary containment tray (no liquid container may go directly on the autoclave shelf). Liquid volumes must not exceed  $\sim 3/4$  of the container capacity to allow for boiling.

**Reagent Tracker**

| Reagent Tracker                                                                              | Quantity  |
|----------------------------------------------------------------------------------------------|-----------|
| <input type="checkbox"/> Item to autoclave (label with name, date, initials)                 | as needed |
| <input type="checkbox"/> Autoclave indicator tape across closure of each item                | 1 strip   |
| <input type="checkbox"/> Loosened lids on bottles / flasks (max $\sim 3/4$ full for liquids) | —         |
| <input type="checkbox"/> Secondary containment tray under all liquid containers              | 1         |

**Loading and running the cycle**

6. Load items loosely, do NOT pack the chamber. Steam must circulate around every item. Do not allow items to touch the chamber walls.
7. Close and seal the autoclave door. Verify the door is fully latched.
8. Select the cycle appropriate for the load (see Notes for cycle selection): (a) Liquids cycle (slow exhaust) for media and aqueous solutions; (b) Gravity / dry goods cycle (fast exhaust) for empty glassware, tips, and instruments; (c) Waste cycle (extended hold) for biohazardous waste bags.

9. Standard parameters: 121 °C, 15 psi, sterilization hold time of 20 min for volumes ≤ 500 mL, 30 min for 500–2000 mL, and 40–60 min for larger volumes or dense loads.
10. Start the cycle and record the start time, contents, and cycle type in the autoclave logbook.

### **Unloading and post-cycle**

11. After the cycle completes, wait until the chamber pressure reads 0 psi and (for liquid cycles) the load temperature is below 80 °C before opening the door.
12. Stand to the side of the door and crack it open ~2 cm to vent residual steam for at least 1 min. Then put on heat-resistant gloves.
13. Remove items slowly. Do not shake or invert liquid containers, superheated liquids can boil over (“flash boil”) when disturbed.
14. Verify sterilization: confirm autoclave tape has darkened uniformly. If any indicator failed, re-autoclave the entire load and notify the lab manager.
15. Allow autoclaved liquids to cool to room temperature before adding heat-labile additives (e.g., antibiotics, amino acid drop-out mixes, glucose for SC media if added post-autoclave).
16. Sign off the logbook with end time, cycle outcome, and tape verification.

### **Notes**

- Heat-labile additives (antibiotics, IPTG, vitamins, glucose in some yeast media) MUST be added after autoclaving via 0.22 µm filter sterilization, never autoclave them.
- Do NOT autoclave: bleach, organic solvents, sealed containers, sharps without a hard-walled container, or any item containing radioactive/toxic chemicals.
- Do NOT autoclave polystyrene or non-autoclave-rated plastics, they will melt. Confirm plastics are polypropylene (PP) or polycarbonate (PC) rated for 121 °C.
- If the autoclave alarms or the cycle aborts, do not open the door until pressure reads 0 psi. Report any malfunction immediately.

### **References**

- Centers for Disease Control and Prevention. *Biosafety in Microbiological and Biomedical Laboratories (BMBL)*, 6th ed. U.S. Department of Health and Human Services, 2020. <https://www.cdc.gov/labs/BMBL.html>
- World Health Organization. *Laboratory Biosafety Manual*, 4th ed. Geneva: WHO, 2020. <https://www.who.int/publications/i/item/9789240011311>
- McMaster University Environmental & Occupational Health Support Services (EOHSS). Biosafety Program Manual. <https://hr.mcmaster.ca/managers/health-safety-well-being/eohss/>
- Sandle, T. (2013). *Sterility, Sterilisation and Sterility Assurance for Pharmaceuticals*. Woodhead Publishing. ISBN 978-1-907568-38-4.